

E-Wave Plane-Wave Scattering by Sinusoidal Open PEC Strips via the Method of Discrete Singularities

Andrii Petryshyn

Department of Theoretical Radio Physics
V. N. Karazin Kharkiv National University
 Kharkiv, Ukraine
 andrii.pe@track360.io

Abstract—We present a full-wave numerical study of plane-wave E-wave (TM) scattering by two-dimensional open sinusoidal perfectly-electric-conductor (PEC) strips. The singular integral equation (SIE) for the induced surface current is discretized by the method of discrete singularities (MDS), a Nystrom-type scheme that exploits Chebyshev quadrature and analytically extracted inverse-square-root edge singularities to achieve spectrally accurate solutions at modest orders ($N = 200$). A flat-strip benchmark confirms correctness: boundary residuals reach machine precision ($\sim 10^{-15}$) and the total scattering width converges monotonically with N . For sinusoidal corrugation the far-field develops lobes at $\varphi_m = \arccos(m\nu)$, matching the Floquet grating prediction for integer order m and spatial frequency ν . A parametric study over $A \in [0.5, 2.5]\lambda$ and $\nu \in [0.10, 0.30]\lambda^{-1}$ shows that A controls the blaze strength (energy fraction in off-specular orders) while ν sets their angular positions. Total scattering width varies by less than 0.5% across all cases, confirming purely angular redistribution of the scattered power.

Index Terms—method of discrete singularities, sinusoidal open strip, plane-wave scattering, grating diffraction, Floquet modes, E-wave, superconducting strip

I. INTRODUCTION

Corrugated metallic surfaces are ubiquitous in microwave engineering. The sinusoidal open strip is a particularly compact example: it serves as a meander-line delay element [10], slow-wave resonator, and unit cell of high-temperature superconducting (HTS) bandpass filters [8], [9]. At microwave frequencies, HTS films satisfy $k\lambda_L \lesssim 10^{-4}$ (London penetration depth $\lambda_L \sim 100$ nm), so the PEC boundary condition is an excellent approximation.

When a plane wave is incident on such a strip, the corrugated surface couples the specularly reflected field into additional propagating orders at angles set by the sinusoidal period. An infinite periodic grating is described analytically by Floquet theory [4], [5], but a finite strip with few periods has significant edge diffraction that demands a full-wave treatment.

Among rigorous 2-D methods for open-arc scattering, the *method of discrete singularities* (MDS) [1], [2] combines analytical edge-singularity extraction with a spectrally accurate Chebyshev Nystrom scheme. The $N \times N$ dense linear system is solved exactly by Gaussian elimination, and boundary

residuals reach machine-precision level. MDS was validated for parabolic multireflector antennas with complex-source-point (CSP) beams [1], [3]; here we apply it to plane-wave excitation and the sinusoidal strip geometry, performing the first systematic grating-diffraction study for this finite corrugated conductor.

II. PROBLEM FORMULATION

A. Boundary-value problem and SIE

We seek the scattered E-wave field U when a 2-D PEC arc Γ is illuminated by a unit-amplitude plane wave

$$U_0(\mathbf{r}) = e^{-ik(x \cos \beta + y \sin \beta)}, \quad (1)$$

with $k = 2\pi/\lambda$, all lengths in wavelengths ($\lambda = 1$, $k = 2\pi$), and $\beta = \pi/2$ (normal incidence, wave propagating in the $+y$ direction). The scattered field satisfies the Helmholtz equation outside Γ , the Sommerfeld radiation condition, local energy finiteness at arc endpoints, and the Dirichlet condition $U|_{\Gamma} = -U_0|_{\Gamma}$.

The single-layer representation of U leads, after parameterizing the arc as $\mathbf{r}(t) = (x(t), y(t))$, $t \in [-1, 1]$, and factoring the inverse-square-root edge singularity into a smooth density $v(t)$ via $j|\mathbf{r}'(t)| = v(t)(1 - t^2)^{-1/2}$, to the first-kind SIE

$$\int_{-1}^1 H_0^{(1)}(kR(t, t_0)) v(t) \frac{dt}{\sqrt{1 - t^2}} = -U_0(t_0), \quad (2)$$

where $R(t, t_0) = |\mathbf{r}(t) - \mathbf{r}(t_0)|$.

B. MDS discretization, geometry, and Floquet orders

Differentiating (2) with respect to t_0 converts the logarithmic kernel singularity to Cauchy type; a supplementary integral condition restores the lost constant. Collocating at $n - 1$ second-kind Chebyshev nodes $\tau_j = \cos(j\pi/n)$, $j = 1, \dots, n - 1$, yields the MDS system [1]

$$\frac{1}{n} \sum_{i=1}^n \left[\frac{v(t_i)}{t_i - \tau_j} + K(t_i, \tau_j) v(t_i) \right] = f(\tau_j), \quad (3)$$

where $t_i = \cos(\frac{(2i-1)\pi}{2n})$ are first-kind Chebyshev nodes, K is the smooth part of the differentiated Hankel kernel, and

$f = -dU_0/dt_0$. The $n \times n$ system is solved by Gaussian elimination. The far-field pattern $\Phi_{sc}(\varphi)$ follows from the discretized single-layer potential asymptotically, and the total 2-D scattering width is

$$\sigma = \frac{1}{\pi} \int_0^{2\pi} |\Phi_{sc}(\varphi)|^2 d\varphi. \quad (4)$$

The sinusoidal strip with $L = 10\lambda$ and phase zero is

$$\Gamma_s = \{(x, y) : x = \frac{L}{2}t, \quad y = A \sin(2\pi\nu x), \quad t \in [-1, 1]\}. \quad (5)$$

For an infinite grating of period $\Lambda = 1/\nu$, normal-incidence Floquet theory gives propagating forward-hemisphere orders at

$$\varphi_m = \arccos(m\nu), \quad |m\nu| < 1. \quad (6)$$

The finite strip ($L\nu$ periods) broadens these into beams centered at (6). With $N = 200$ nodes the worst-case arc length ($A = 2.5\lambda$, $\nu = 0.30$: $\approx 48\lambda$) provides ≈ 4.2 nodes per wavelength, sufficient for spectral accuracy.

III. NUMERICAL RESULTS

A. Flat strip benchmark

Setting $A = 0$ gives a flat horizontal strip of length $L = 10\lambda$. Near-field and far-field results for normal incidence are shown in Fig. 1; the boundary residual is 2.12×10^{-15} (machine precision). The far-field pattern exhibits the expected two-lobe structure: the 90° lobe corresponds to forward diffraction (shadow region), and the 270° lobe to backward specular reflection, with weaker edge-diffraction contributions at 0° and 180° .

Table I shows N-convergence of σ . The sequence is monotone and the residual stays below 3×10^{-15} throughout, confirming that the inverse-square-root edge singularity is correctly accounted for and the discrete system is solved to machine precision.

TABLE I
N-CONVERGENCE: TOTAL SCATTERING WIDTH σ FOR THE FLAT STRIP
($A = 0$, $L = 10\lambda$, $\beta = \pi/2$).

N	σ (λ)	Rel. change	Residual
32	31.866	—	7.2×10^{-16}
64	35.954	1.3×10^{-1}	8.5×10^{-16}
100	37.710	4.9×10^{-2}	2.3×10^{-15}
150	38.765	2.8×10^{-2}	1.7×10^{-15}
200	39.243	1.2×10^{-2}	2.1×10^{-15}

B. Sinusoidal strip: Floquet-order identification

For $A = 1.5\lambda$, $\nu = 0.20\lambda^{-1}$ ($L\nu = 2$ periods) the boundary residual is 2.92×10^{-15} . In the near-field [Fig. 2(a)], the corrugation disrupts the regular standing-wave pattern of the flat strip, producing complex interference structures above and below the arc.

The far-field [Fig. 2(b)] reveals the key result: beyond the specular lobes at 90° and 270° , pronounced peaks appear at 78.5° , 101.5° , 66.4° , and 113.6° , matching the Floquet

prediction (6) for $m = \pm 1$ and $m = \pm 2$ with $\nu = 0.20\lambda^{-1}$. The backward hemisphere shows a mirror structure. Although the finite strip ($L\nu = 2$ periods) broadens each lobe, the peak positions unambiguously confirm the Floquet angles.

C. Parametric pattern evolution

Figure 3 summarizes two parametric sweeps over the forward hemisphere (0° – 180°).

Amplitude dependence [Fig. 3(a)]: For the flat strip ($A = 0$) a single dominant lobe sits at 90° . As A increases at fixed $\nu = 0.20\lambda^{-1}$, lobes grow monotonically at the $m = \pm 1$ Floquet angles (78.5° , 101.5°), while the specular peak broadens. At $A = 2.5\lambda$ the first-order lobes are within a few decibels of the specular peak, indicating substantial power transfer to off-normal directions — the finite-strip analog of diffraction-grating blaze.

Frequency dependence [Fig. 3(b)]: With $A = 1.5\lambda$ fixed, increasing ν shifts the $m = +1$ lobe from 84.3° ($\nu = 0.10$) to 72.5° ($\nu = 0.30$) following (6) exactly. At $\nu = 0.25$ – 0.30 the $m = \pm 2$ lobes enter the visible range. Across all 25 configurations the total scattering width σ varies by less than 0.2λ ($\approx 0.5\%$), confirming that the sinusoidal corrugation redistributes rather than increases the total scattered power.

IV. CONCLUSION

The method of discrete singularities was applied to E-wave plane-wave scattering by sinusoidal open PEC strips. The MDS solver was validated on a flat-strip benchmark: boundary residuals reached machine precision and σ converged monotonically with N . For the sinusoidal strip, far-field lobes at the Floquet angles $\varphi_m = \arccos(m\nu)$ were confirmed numerically for the first time in the finite-strip setting. The parametric study identifies two independent design knobs: corrugation amplitude A controls the blaze strength (fraction of power in off-specular orders), and spatial frequency ν sets the grating-order angles. Total scattering width remains practically constant ($\lesssim 0.5\%$), confirming purely angular redistribution. For superconducting meander-line devices these findings imply that the corrugation profile controls the far-field scattering signature without altering the total radiated power, enabling shape-based crosstalk management in densely integrated microwave circuits. Planned extensions include H-wave polarization, oblique incidence, and impedance boundary conditions for resistive or finite-conductivity layers.

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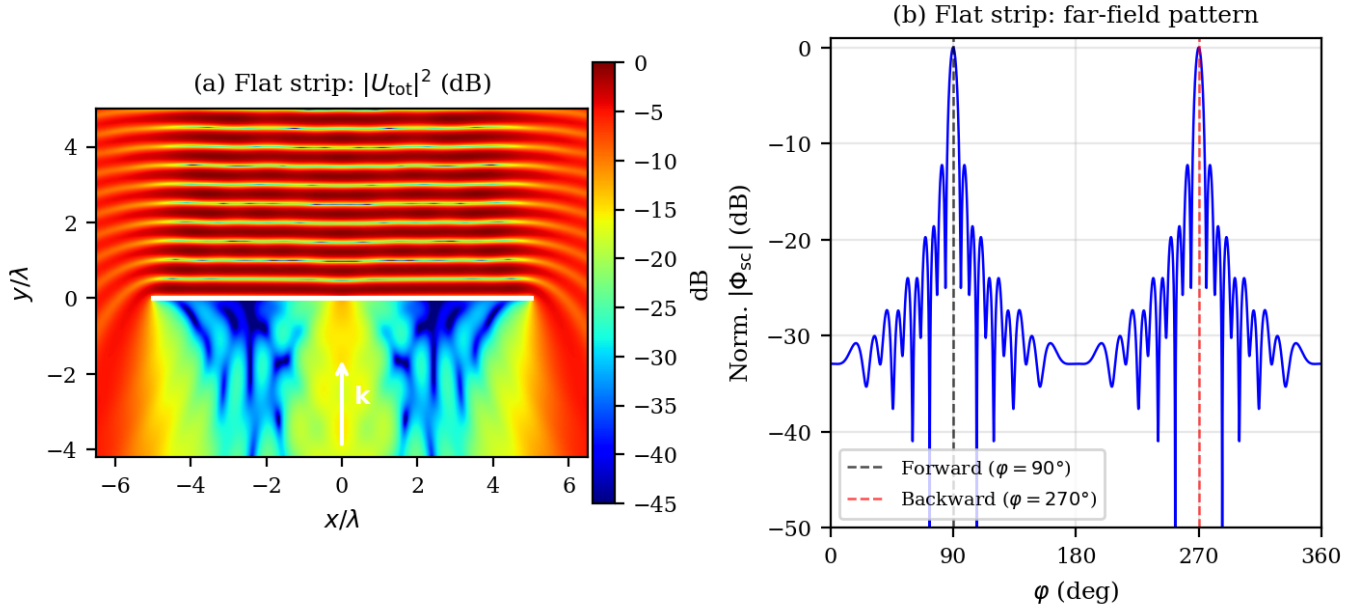


Fig. 1. Flat strip ($A = 0$, $L = 10\lambda$, $N = 200$, $\beta = \pi/2$). **(a)** Total near-field intensity $|U_{\text{tot}}|^2$ (dB, normalized to peak); the arrow marks the incident-wave direction. A standing-wave pattern is visible below the strip; the shadow region above shows reduced amplitude. **(b)** Normalized far-field scattered amplitude $|\Phi_{\text{sc}}|$ (dB) vs. φ : forward diffraction at 90° , specular reflection at 270° , edge-diffraction contributions near 0° and 180° .

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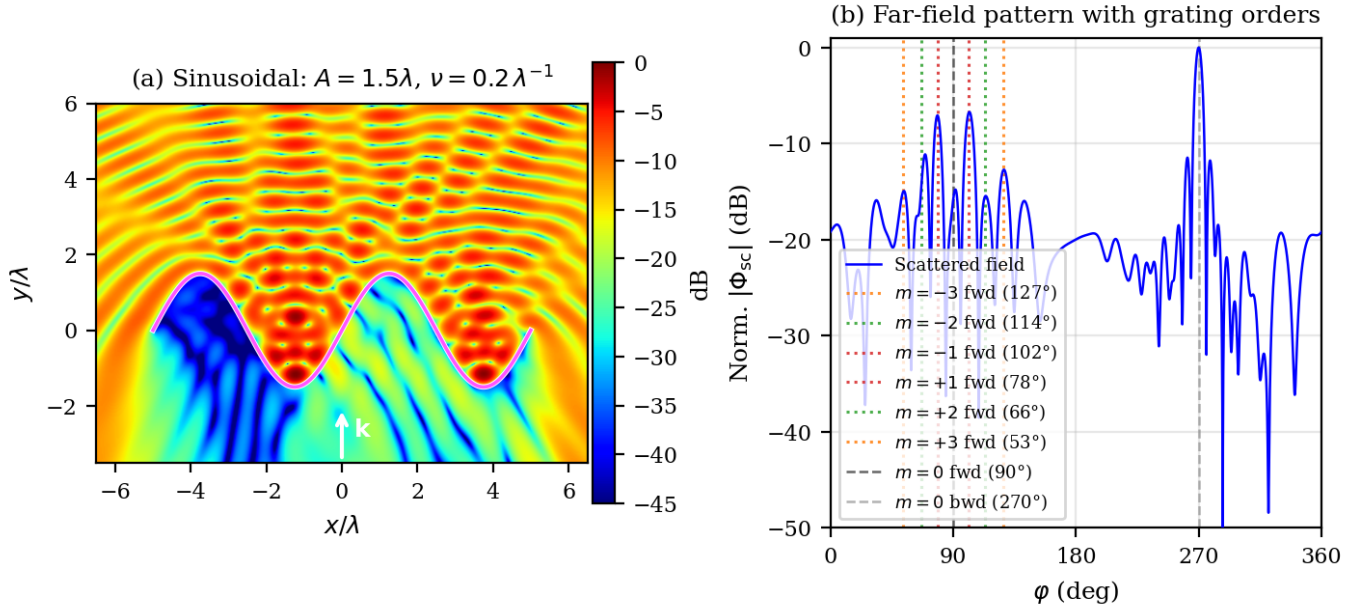


Fig. 2. Sinusoidal strip ($A = 1.5\lambda$, $\nu = 0.20\lambda^{-1}$, $L = 10\lambda$, $N = 200$). (a) Total near-field intensity; strip profile in magenta. (b) Normalized far-field amplitude (dB); vertical markers show predicted Floquet angles: $m = 0$ at 90° ; $m = \pm 1$ at $78.5^\circ/101.5^\circ$; $m = \pm 2$ at $66.4^\circ/113.6^\circ$.

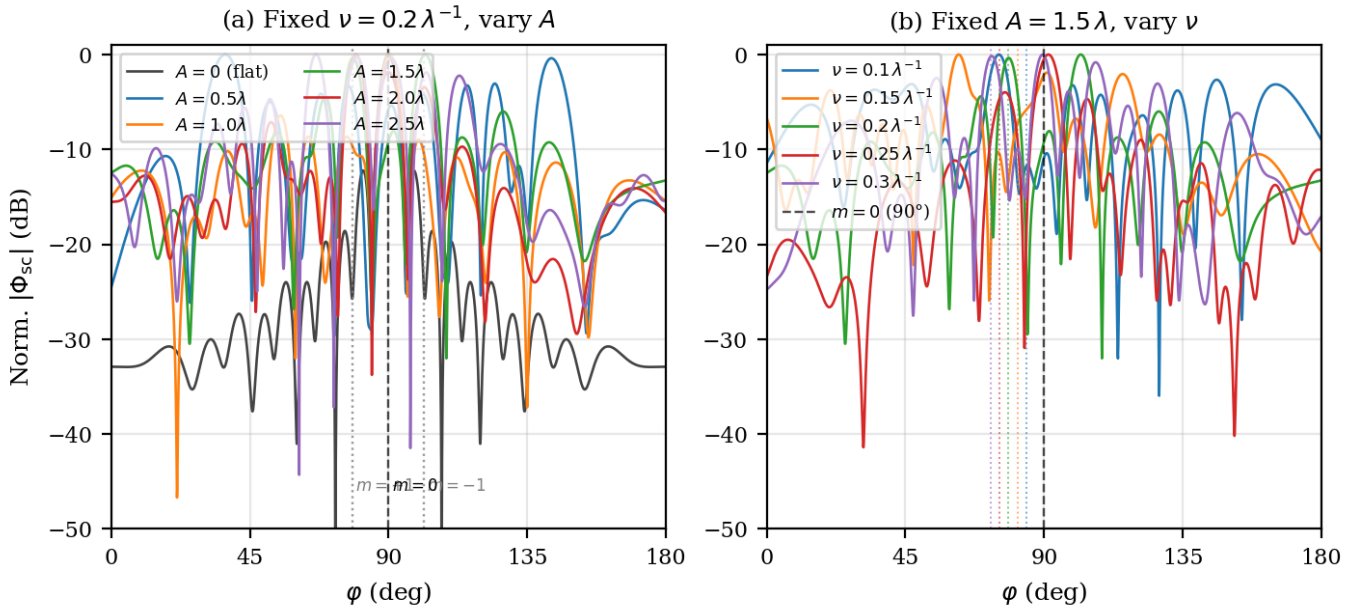


Fig. 3. Forward-hemisphere (0° – 180°) normalized far-field amplitude (dB) illustrating grating-order development. (a) Fixed $\nu = 0.20\lambda^{-1}$, A varied from 0 to 2.5λ ; dotted verticals mark $m = \pm 1$ Floquet angles (78.5° and 101.5°). (b) Fixed $A = 1.5\lambda$, ν varied; dotted verticals mark the $m = +1$ angle for each ν ($84.3^\circ \rightarrow 72.5^\circ$ as ν increases).